Volatility Forecasting with GARCH

Gaussian GARCH(1,1) had the lowest forecast loss across 1,760 trading days from 2019 to 2025. Its mean squared error was 13% lower than EWMA's and 27% lower than rolling variance's. The two GARCH specifications finished much closer together.

Research question

Large daily stock returns often cluster in time. GARCH estimates how strongly return shocks affect variance and how long their effects last. Does this improve next-day variance forecasts over rolling variance or an exponentially weighted moving average (EWMA)? We test this on SPY, an ETF tracking the S&P 500.

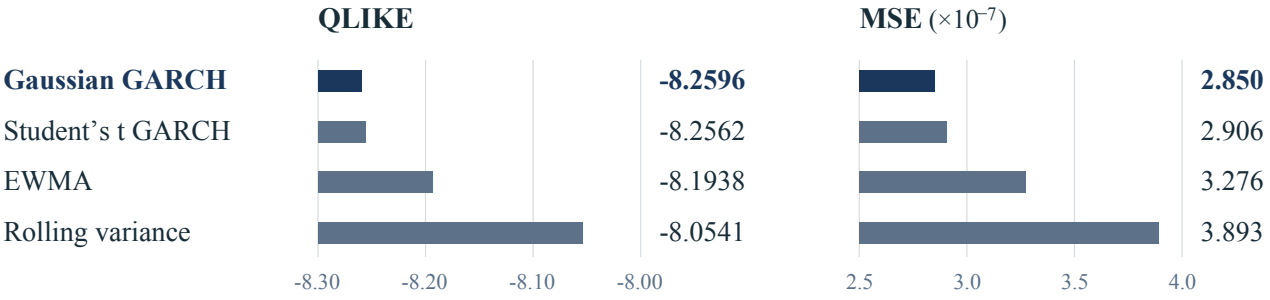
Methodology

We compute daily log returns from SPY adjusted closing prices. We implement the variance recursions in Python and estimate Gaussian and standardized Student's t GARCH(1,1) by maximum likelihood. Both use the previous squared return shock and the previous variance forecast.

We train on 2010–2015 and select the rolling window (60 trading days) and EWMA decay (0.94) on 2016–2018. We refit GARCH on 2010–2018, then hold coefficients fixed throughout testing. Each forecast uses only past returns. We score it against the next day's squared deviation from the pretest mean return.

Results

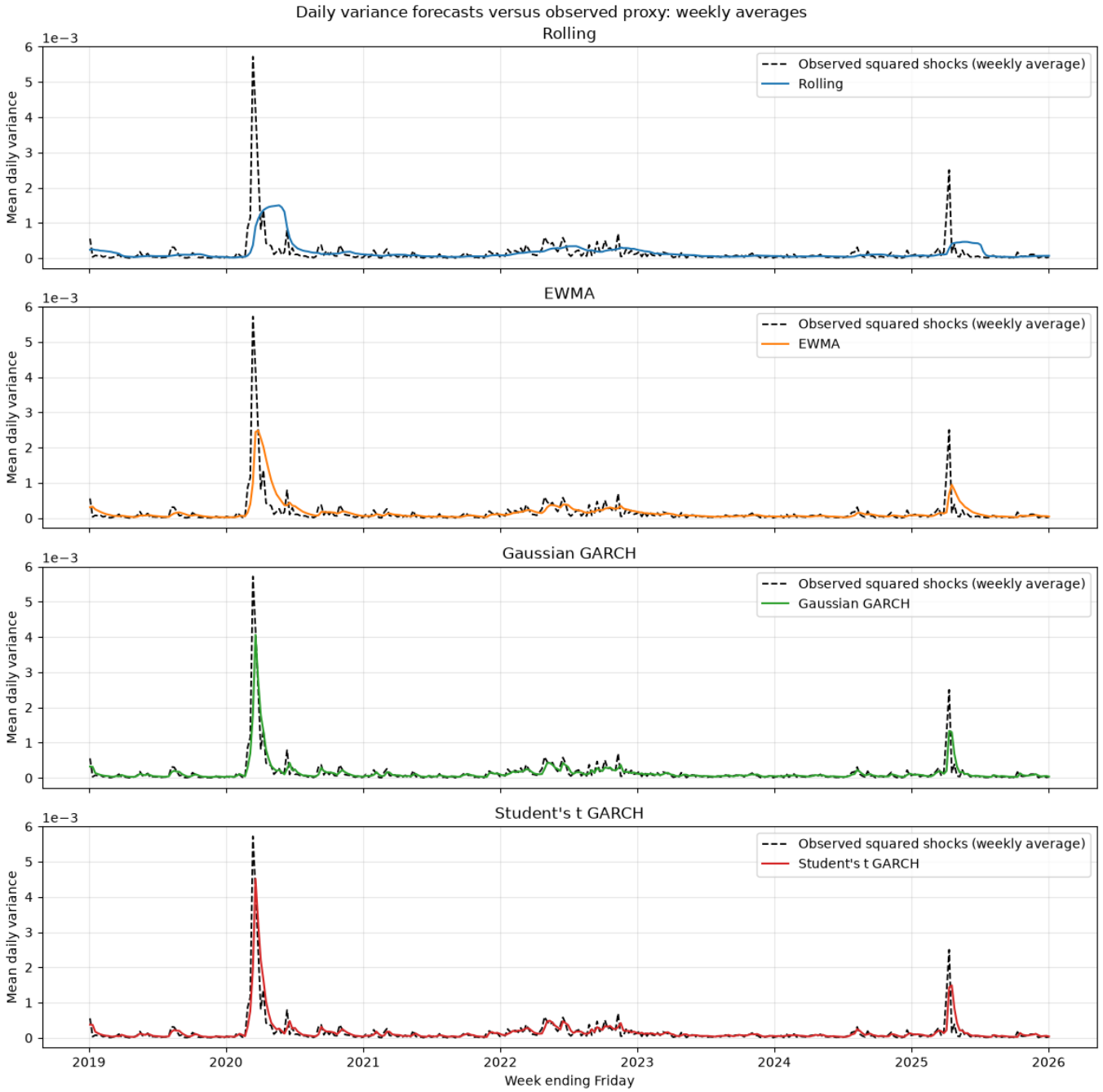
QLIKE and mean squared error (MSE) rank the models in the same order. Lower scores are better. Both axes are zoomed. QLIKE starts at -8.30, and MSE starts at 2.5 in the displayed units.



GARCH estimates separate weights for new shocks and past variance, and its forecasts revert toward a long-run level. These features may help explain its advantage over the benchmarks. Student's t GARCH had about 2% higher MSE than Gaussian GARCH. In the weekly-average figure on the next page, GARCH also tracks the observed proxy more closely than rolling variance and EWMA, particularly during volatility spikes.

Limitations

Daily squared shocks are just a noisy proxy for unobserved variance. The evidence covers one ETF and one test period. We did not test the loss differences for statistical significance. We added the Student's t specification after inspecting earlier test results, so its comparison is exploratory.



Each panel compares one model with the same observed proxy. Daily variance forecasts and observed squared return shocks are averaged over the same days in each week ending Friday. Weekly averaging reduces visual noise. These are averages of daily variances, not variances of weekly returns.